

Interactive Research Artifacts

Interactive Technologies as Tools for Knowledge Creation

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ABSTRACT

New technologies and digitization have the potential to vastly alter our knowledge infrastructures. Specifically, this work focuses on the effects of interactive technologies on research practices, referred to as “interactive research artifacts.” Current research investigates the communicative affordances of such technologies, but there exists minimal work which critically examines the creative ways scholars are engaging with these artifacts. Through in-depth interviews with 14 scholars, and Design Studies literature, this work arrives at an understanding of interactive research artifacts as creative knowledge creation tools, rather than simply communicative tools. As such, to design for a future where interactive research artifacts become commonly used scholarly tools, an in-depth understanding of the ways these artifacts are used as knowledge creation tools is critical.

CCS CONCEPTS

• **Human-centered computing** → Interaction design theory, concepts and paradigms; Human computer interaction (HCI).

KEYWORDS

Interactive artifacts, knowledge creation, research practices

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1 INTRODUCTION

New forms of technologies and computation are becoming intertwined with our notions of society and knowledge, yet at the same time, digital innovation is paving the way for new possibilities for our global knowledge infrastructures. Novel technologies are enabling new modes of collaboration, openness, and research through diverse forms of digital scholarship [1, 18]. With this increasing emphasis on digitalization, more work is required to conceptualize and support the ways we construct scholarship. However, to better design for these futures, we must understand exactly how these technologies are created and used currently. Despite the recent uptick in adoption, there is minimal research which critically examines the applied use of interactive technologies in research practices. As

such, this research explores the current use of interactive technologies in scholarly practices. I seek to conceptualize and further our understandings of these technologies as “interactive research artifacts” and knowledge creation tools.

This work uses the term *interactive research artifacts* to categorize various dimension of interactive digital artifacts used in research practices. This term is purposefully broad, as there is minimal literature which currently conceptualizes these artifacts. The aim of this work is to build towards an understanding of interactive research artifacts as understood by current users. Examples range from one-off websites, such as “Six Degrees of Francis Bacon”, a Digital Humanities project, (Figure 1) and “Project OKAI”, an “interactive introduction” to topics in AI, [22]; interactive textbooks; web-based programming notebooks, notably Jupyter, Nextjournal, and Observable; to full periodical journals, such as *Distill* and *Vectors*. These span multiple disciplines, from Digital Humanities to Computer Science. For many, these interactive research artifacts offer explorations of data otherwise impossible with traditional methods. In “Six Degrees of Francis Bacon” [23], users are visually able to examine relationships by directly engaging with the network diagram. For *Distill*, interactive research artifacts often illustrated complex topics. These artifacts leverage previously studied affordances of interactivity to improve engagement between humans and computers [17]. Many refer to their work as “explorable explanations,” others use interactive articles [8], multimedia artifacts [10], and more. Despite differences in terminology and disciplines, these examples are united by their understanding of interactive media as a critical component of scholarly practice. Therefore, this work aims to build a foundation for conceptualizing this phenomenon. Based off insights from current users, an initial working definitions of interactive research artifacts is proposed at the end of this paper.

There is a breadth of research acknowledging the changing relationships between interactive technologies and knowledge infrastructures, defined as “robust networks of people, artifacts, and institutions that generate, share, and maintain specific knowledge about the human and natural worlds” by [4]. This includes work on interactive learning technologies, such as interactive textbooks and classroom settings [14, 16, 18], interactive data visualizations, particularly in computational notebooks [7, 13, 20, 21], and interactive journalism and storytelling [5, 8, 12]. While these works provide important insights into the varying ways interactive technologies transform our knowledge infrastructures more broadly, we cannot safely transfer insights from interactive textbooks and data journalism to scholarly research, and as such they do not provide us with contextual understandings of the ways these technologies transform such research practices.

Some work has begun to investigate of interactive artifacts, including online movements such as Explorable Explanations [19] and research on communicating with these artifacts [3, 8]. This

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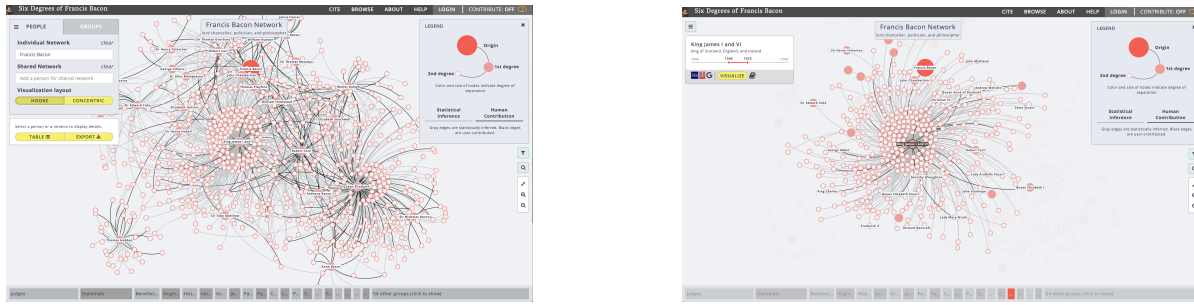


Figure 1: “Six Degrees of Francis Bacon” (interactive social network of Francis Bacon). This shows how users can select individuals or specific connections to better visualize the social networks connected to Francis Bacon[23]

Table 1: Participant Demographics

Participant	Profession	Discipline	Gender
P1	Co-founder of computational notebook company, Professor	Computer Science	Male
P2	Designer	Design	Female
P3	Professor	Art History	Male
P4	Entrepreneur (technology)	Technology, Journalism	Female
P5	Research support team member	Humanities	Male
P6	Research support team member	Humanities	Female
P7	Research support team member	Humanities	Female
P8	Co-founder of computational notebook company	Computer Science	Male
P9	Professor	Human Computer Interaction	Male
P10	Entrepreneur (educational technology)	Educational Technology	Female
P11	Professor	Art History	Male
P12	Educational technology consultant	Educational Technology	Female
P13	Professor, journal co-editor	Education	Male
P14	Interaction designer	Design	Male

provides valuable insights into the changing nature of scholarly communication [9, 11]. However, there is minimal work which critically explores the use of interactive research artifacts in practice, leaving us without an understanding of how scholars conceptualize them. This leads to this work’s research question:

RQ: How do scholars currently use and conceptualize interactive research artifacts?

2 METHODS

This paper reports on 12 online in-depth interviews with 14 participants, in which their current practices regarding interactive research artifacts were explored. All participants identified as actively creating or using interactive research artifacts. Three participants on the same research support team requested to do a group interview; they felt their collective knowledge was more relevant rather than their individual knowledge, as their work was collaborative.

For the scope of this research, “scholars” simply refers to anyone who identifies as engaging with scholarship. As discussed, there is no comprehensive understanding of interactive research artifacts. Therefore, a call was sent out for those using “interactive research artifacts,” described as using interactive technologies in scholarly practices. Rather than presenting participants with a definition

of this, this work allowed participants to identify themselves as creators and/or users of these artifacts. Directly engaging through semi-structured interviews provides insights into participants’ relationships between interactive research artifacts and their research processes. Semi-structured interviews are a conversation in which knowledge is constructed and expose deeper meanings regarding interviewees’ perspectives.

Interviewees were identified in varying fields, ranging from Human-Computer Interaction (HCI), Computer Science, Digital Humanities, and Education, whose commonality is their interest in exploring their scholarship via non-traditional means by leveraging interactive research artifacts. After interviewing the initial participants, further interviewees were identified via snowball sampling. All interviewees (see Table 1) met the following criteria: they 1) express interest in interactive research artifacts of various forms 2) have experience using interactive artifacts in scholarly contexts and 3) would voluntarily participate in an interview. No incentives were provided for participation.

3 FINDINGS

During the advent of computer programs, it was argued that computers do not simply duplicate interactions which occur offline, but

rather, their unique affordances can encourage different ways of processing and analyzing information [15]. The same conclusions can be said now – the experiences of these interviewees indicate that interactive research artifacts are providing scholars with different ways to approach their scholarship.

The interviews illustrate a shared understanding of interactive research artifacts as tools for knowledge creation. As P3 described, they “*bring about the dialogues of knowledge creation.*” These findings suggest there are two purposes of said artifacts, to: (1) improve scholarship quality and (2) enable play with data. Two types of these artifacts were detailed by interviewees, which are interactive research artifacts as (1) communicative tools and (2) research process tools.

3.1 Using Interactive Research Artifacts to Improve Research Quality

Most interviewees explicitly detailed how interactive research artifacts often serve a hidden purpose: they are tools for uncovering new insights or inconsistencies in the underlying data that is often missed when using traditional, non-interactive artifacts. P2, an interaction designer who frequently collaborates with scientists to create interactive research artifacts, illustrated this point by offering an example whereby engaging with an interactive data visualization allowed them to discover errors in their climate model data. He believes interactive research artifacts allow researchers to look at their data from different perspectives, enhancing the caliber of the scholarship.

The research support team at a UK university provided a slightly different insight: instead of being used to locate errors in data, they leveraged interactive research artifacts to shine light on differing insights. When asked what aspect of interactive research artifacts interests her, P6 replied, “*it’s the new perspectives that they may offer on research data. You can use it to discover new things about the data or you can explore the data in ways you might not have expected otherwise.*” Interactive research artifacts are used by scholars to locate areas that warrant further research and investigation. The rest of the team agreed. To them, these artifacts are wielded as discovery tools, which serve to improve the quality of scholarship by providing a new perspective to uncover insights that might not be considered otherwise.

By providing alternative, non-linear modes of exploring knowledge, interactive research artifacts allow P11, an art historian focused on Leonardo Da Vinci, to produce a caliber of scholarship not quite possible by traditional means: “*you could explore things in a rather Leonardo-esque way. You’re not just looking at one compartmentalized bit of Leonardo’s knowledge classified according to modern taxonomies of knowledge.*” Likewise, some scholars use interactive artifacts to pinpoint areas of their research process needing improvement. P13 explains how interactive research artifacts provide a “*nudge that changes pedagogy.*” Specifically, his team uses these artifacts to reflect on their knowledge creation processes and locate areas which could use more diversity in thought, method, and communication. In this way, interactive research artifacts are not only a tool for discovering new information; they serve as a reflexive tool.

For many, interactive research artifacts are tools which enhance scholarly inquiry. As P1 described: “*computers are a thinking partner.*” Interactive research artifacts facilitate a new dynamic for knowledge creation. They are used to uncover new insights and perspectives, reflect on the work conducted, speed up the research process, and connect information in ways not possible by the affordances of traditional artifacts. They not only serve as a tool to improve the process of knowledge creation and development of scholarship, but they also provide varying platforms which allow scholarship to be expressed in a manner befitting the subject.

3.2 Playing with Data to Uncover New Knowledge

Most participants directly addressed the role of play in their research practices. These artifacts offer explorations not possible with traditional artifacts, enabling the discovery of new knowledge through personal experimentation and exploration. Four described their work as “Explorable Explanations:” “*These kinds of explorable explanations and papers can sit somewhere in between your findings and your raw data, allowing an extra level of exploration and querying in respect to individual context and viewpoints*” [P9].

P12 details how traditional artifacts are “linear”, which limits the ways she can engage with them. Interactive research artifacts allow for playful exploration, which “*appeals to different users’ modalities.*” P10 took this further and understood interactive research artifacts to be tools to get others to play with data, fostering outside exploration and therefore generating new strands of knowledge she might have missed. P4 likewise emphasized exploration by diverse users to “*break out and create something that was different.*”

Data does not only refer to quantitative data – some participants described how remixing existing interactive research artifacts fosters dialogue between the users and the content itself. Not only does this element of interactive research artifacts allow users to leverage existing scholarship, but it also encourages them to conceptualize information differently by providing a multiplicity of options. As P8 described, you can remix an artifact from another topic or discipline with your own scholarship. This may enable the discovery of new insights otherwise obscured. For these participants, knowledge creation is enabled through creative exploration and playful experimentation.

3.3 Two Classifications of Interactive Research Artifacts

As suspected from previous literature in this space, interactive research artifacts can act as communicative tools for scholarly knowledge. While only some participants detailed their understandings of interactive research artifacts to include this communicative tool classification; all discussed, both implicitly and explicitly, the ways interactive research artifacts alter knowledge creation practices during the research stage, rather than only the output stage.

Participants detailing the impact of these artifacts on scholarship quality primarily discussed this relating to research process tools. They used these to interrogate data, uncover new insights, and spot errors in their work. However, some described these artifacts as communicative tools which enabled their work to be expressed in a more “*authentic manner*” [P9]. “*You can shape your argument*

differently than how you can with just a static paper” [P1]. When understanding these artifacts as communicative tools, focus shifted to the structure of the knowledge expressed. When conceptualized as research process tools, scholars perceived a transformation in their knowledge creation processes.

The playful use-case of interactive research artifacts was prominent in both artifact classifications. Work regarding Explorable Explanations begins to address the role of play in these communicative artifacts [3]. Here, “readers” (those engaging with the research who are not authors) can engage with the scholars’ outputs in novel ways, fostering unique interpretations of the research, and generating new knowledge. However, when used as research process tools, the act of play occurred in initial research stages. These playful interactions often resulted in new insights.

Many participants, while describing their views on interactive research artifacts, also detailed challenges they faced. When used as communicative tools, the scholars detailed problems plaguing other types of new research movements, such as access, incentive structures, and peer review, of which there is substantial literature on [1, 2, 6]. However, the challenges for the artifacts as research process tools differed. Predominately, there was a lack of awareness of best practices for creating said artifacts. Scholars described being unaware of how to create these artifacts, what design processes to follow, and what tools were best to use. Time and lack of necessary skills were the biggest barrier of entry.

4 CONCLUSION AND FUTURE WORK

Current conceptualizations of interactive research artifacts have not yet holistically examined them through the lens of knowledge creation processes. While findings of this work support previous studies which examine the scholarly communication impact of these artifacts, this work takes this a step further. Interactive research artifacts are understood to be tools for knowledge creation, which is enabled by improving scholarship quality and promoting playful exploration. As such, this work suggests generalizing these as broadly knowledge creation tools. I posit an initial working definition of interactive research artifacts as “digital data and scholarly evidence which encourage user engagement in the creative process of knowledge creation.” Conceptualizing interactive research artifacts as tools for exploring scholarship as a means of knowledge creation allows for a more nuanced and detailed understanding of their possibilities. Interactive research artifacts create new framings of inquiries and exists within a constellation of other research artifacts which seek to tackle this space. To continue to improve our knowledge infrastructures, we must develop understandings of the various ways this diversity of research artifacts can tangibly impact them.

More work must be done to critically examine the creation and use of interactive research artifacts in detail so that we may design for the future. Future work should continue to engage with diverse groups to further our understanding of this phenomenon. Despite increased focus on interactive research artifacts, our lack of critical understanding of them means we do not know how to properly facilitate their design. *“It feels like we haven’t quite solved the problem yet. We have various ideas for part of it. When a researcher comes to us and says, ‘I’m building a database of 14th century manuscripts,*

I would like to make this available online, how do I go about doing that?’ I would like there to be a simple clear answer. By being able to provide a more customized solution where that’s more appropriate, we can provide people with what they want rather than fitting what they want to what we can provide” [P7]. As such, this work posits that the next question to be answered should be: how can we best facilitate the creation of these interactive research artifacts?

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